

Worksheet

1. Does the sequence $\{\sqrt[n]{n}\}$ converge? If so, what is the limit? If not, why not?

2. Find $\lim_{n \rightarrow \infty} \left(1 - \frac{3}{n}\right)^{2n}$.

3. Find $\lim_{n \rightarrow \infty} 3^{1/n}$.

4. Find $\lim_{n \rightarrow \infty} r^n$, where r is some real number. Does your answer depend on what r is? If so, explain how. If not, explain why.

5. Suppose $\{a_n\}$ is a sequence of positive numbers and $a_n < \frac{1}{n^2}$. Must $\{a_n\}$ converge? Why or why not?